Trajectory Tracking of a Two-DOF Planar Robotic Arm

Kinematics | Joint-space P / PI / PID control | Simscape Multibody

1. Project overview

This project studies a two-revolute-joint robotic manipulator in MATLAB/Simulink and Simscape Multibody. The goal is to follow a prescribed circular end-effector trajectory by converting Cartesian references into joint-angle commands and applying torque feedback control.

2. Reference motion and geometry

The first and second links are 40 cm and 30 cm long. The target is a circle with center (50, 20) cm, radius 10 cm, and period 5 s. The reference components are:

$$x_d(t) = 50 + 10\cos(2\pi t / 5) \quad [cm]$$

$$y_d(t) = 20 + 10\sin(2\pi t / 5) \quad [cm]$$

3. Implementation and experiments

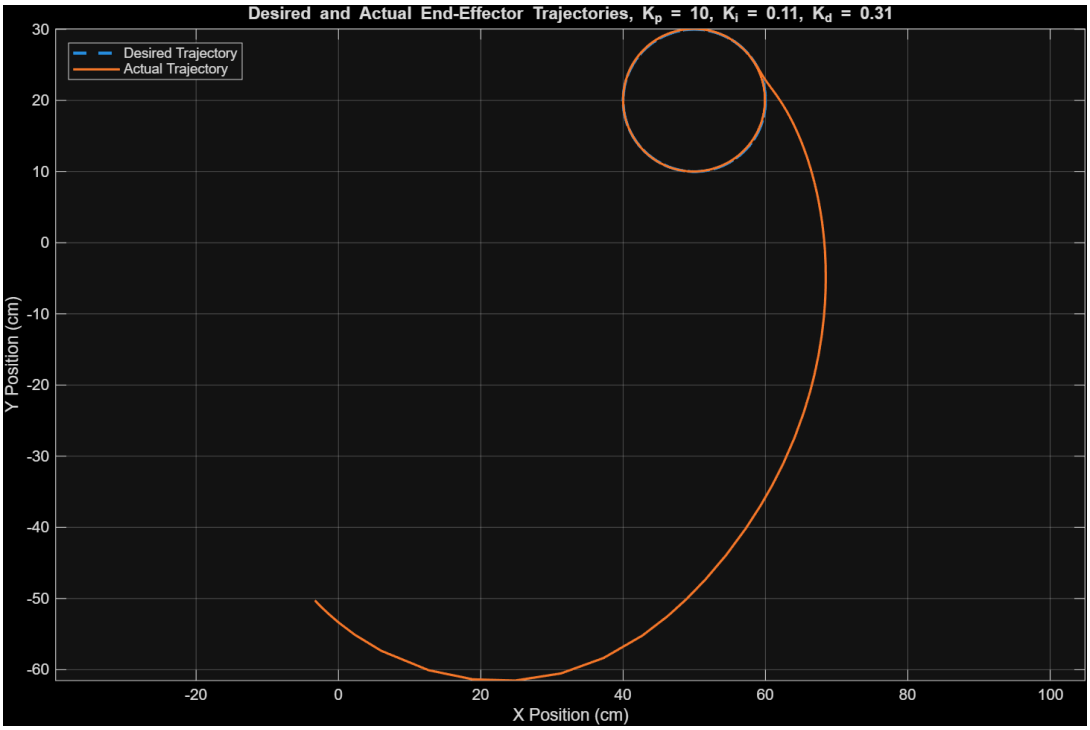
- Reference generation: Simulink sine-wave blocks generate the periodic x-y motion.
- Kinematics: geometric inverse kinematics produces desired joint angles; forward kinematics reconstructs the end-effector position.
- Mechanical model: a CAD-derived, two-link Simscape Multibody arm provides joint dynamics.
- Feedback control: each joint has a controller driven by its angular tracking error; the outputs command joint torques.
- Gain studies: individual P, I, and D changes are evaluated against tracking paths and torque transients.

4. Controller settings

Sweep / stage	Parameter settings
P study	Kp = 0.5, 1, 2, 5, 10 (Ki = Kd = 0)
PI study	Kp = 1, Ki = 0.1, 0.5, 1, 2, 5
Derivative study	Kp = Ki = 1, Kd = 0.1, 0.5, 1, 1.5, 2
Transferred PD	Kp = 4.15, Ki = 0, Kd = 0.15
Final tuned PID	Kp = 10, Ki = 0.11, Kd = 0.31

Tracking performance and observations

Reference vs. actual end-effector trajectory - final tuned PID

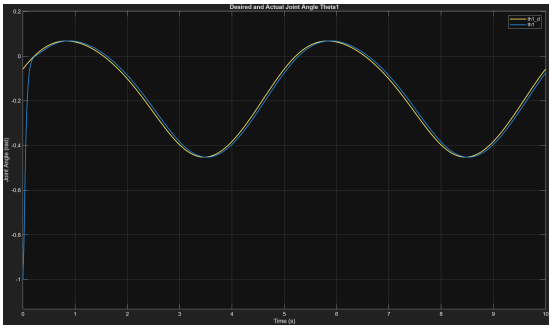


Final simulation: Kp = 10, Ki = 0.11, Kd = 0.31; Cartesian position in cm.

Key findings

- The controller brings the arm toward the reference circle after an initial transient, then closely follows the periodic path.
- Raising Kp generally improves response speed and tracking. Excessive integral or derivative gains amplify oscillations or initial torque spikes.
- The final reported mean Cartesian tracking error is 1.2754 cm, computed over the full simulation including its initial transient.

Joint-angle tracking - final controller



Desired and actual angle of the first joint.

Limitations and provenance

These are pre-existing simulation results, not hardware experiments. A fixed-gain PID does not explicitly compensate for manipulator coupling or changing payloads. No new MATLAB execution or independent numerical re-validation was performed while preparing this short publication report.

Source: Submitted 44-page Part II report and accompanying MATLAB/Simulink files.